🔔 Automatic Doorbell Using IR Sensor — Project Theory

🌟 Project Aim:

To create an automatic doorbell system that rings whenever a person comes near the door using an IR sensor and plays a recorded voice/message using an ISD1820 voice module.

🧠 Main Idea:

We are using a sensor that can detect people without physical contact and a voice module that can play a sound automatically. This setup replaces the need for pressing a doorbell button.

🧩 Components Used:

1. **IR Sensor** – Detects when a person is near the door.
2. **Arduino UNO** – Acts like the brain of the system.
3. **ISD1820 Voice Module** – Plays a recorded bell sound or voice.
4. **Connecting Wires & Power Source**

🛠️ How It Works — Step by Step:

Step 1: IR Sensor Detection

- The IR sensor emits infrared light.
- When a person comes near, this light reflects off their body.
- The sensor detects this reflection and sends a signal to the Arduino.

Physics Connection:
This is similar to the reflection of light — just like our eyes detect light reflected from objects, the sensor detects reflected infrared light.

Step 2: Arduino Processing

- The Arduino receives the signal.
- It tells the voice module to play the sound.
- It waits for 10 seconds before allowing the bell to ring again.
- If the visitor is still standing after 5 seconds and no one opens the door, it rings again to alert people inside.

Physics/Logic Connection:
Arduino uses logic (like if-else) and timers to control actions. It counts time (delay) and follows conditions.

Step 3: Voice Module Output

- The ISD1820 has a recorded sound.
- When activated, it plays this sound through a small speaker.

Physics Connection:
The module turns electric signals into vibrations (sound) that we hear — same principle as a speaker.

✅ **Final Outcome:**

- Person arrives → Bell rings.
- Person keeps standing → Bell rings again after 5 seconds.
- Another person comes → Bell rings again, even if 10 seconds haven't passed.

Concept	How It's Used
Reflection of Light	IR sensor uses reflected light to detect
Sound Production	Speaker converts electrical energy to sound
Electric Current	Powers all components
Logic and Time Delay	Arduino uses code logic and timers

🧪 **Real-Life Applications:**

- **Touchless Doorbells:** In hospitals or during pandemics.
- **Smart Home Entry Systems**
- **Shops/Offices:** Automated visitor alerts.